

# LAB PROGRAM 1: Build a Docker Container from a Custom Dockerfile

Project Structure:

Program-1/

-----> Dockerfile

-----> requirements.txt

-----> [app.py](#)

Create a Dockerfile, requirements.txt and app.py files with the following content

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ ls
app.py  Dockerfile  requirements.txt
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ cat app.py
from flask import Flask
app = Flask(__name__)
@app.route("/")
def hello():
    return "hello world"
if __name__ == "__main__":
    app.run(host="0.0.0.0",port=5000)
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ cat requirements.txt
Flask
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ cat Docekrfile
cat: Docekrfile: No such file or directory
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ cat Dockerfile
#use a slim python base image
FROM python:3.9-slim
#set workdir inside container
WORKDIR /app
#copy the requirements file and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000
CMD ["flask","run"]
```

Execute the Docker Build command on the Terminal to build the Docker image

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ sudo docker build -t prog1 .
[sudo] password for 1RV24MC025_CHANDAN_K:
[+] Building 2.4s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 378B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim  2.2s
=> [auth] library/python:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338 0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 93B                                       0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> CACHED [5/5] COPY . .                                           0.0s
=> exporting to image                                              0.0s
=> => exporting layers                                              0.0s
=> => writing image sha256:4cf0767d901b17759c70865f9b2252311016e92c7240d 0.0s
=> => naming to docker.io/library/prog1                             0.0s
```

Execute the Docker run command specifying the Port Number for the container.

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ sudo docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
prog5-jenkins        latest             9b0beb1de723       4 weeks ago        925MB
prog3                latest             9663bfdfd239       4 weeks ago        1.13GB
chandank30/prog6     latest             3f0e53b14715       4 weeks ago        1.09GB
prog6                latest             3f0e53b14715       4 weeks ago        1.09GB
ghcr.io/chandank30/prog6 latest             3f0e53b14715       4 weeks ago        1.09GB
image2               latest             bc3115a9bf0a       6 weeks ago        8.32MB
image1               latest             c4b794f7d0ec       6 weeks ago        8.32MB
prog2                latest             c8e272046bc2       6 weeks ago        138MB
prog1                latest             4cf0767d901b       6 weeks ago        133MB

1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog1$ sudo docker run prog1
* Serving Flask app 'app.py'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [05/Jan/2026 09:36:35] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [05/Jan/2026 09:36:35] "GET /favicon.ico HTTP/1.1" 404 -
```

Test the Container by verifying the localhost details on web browser, the text →  
Hello Docker! Will be displayed.

